Autoencoder Market Models for Interest Rates

Replication of Sokol (2022) & CVAE-Driven Trading Signals

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Topics in Derivative Pricing
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Roadmap for the next 20 minutes

Act 1 - The problem. Why compress the yield curve, and why Nelson-Siegel is not the last word.

Act 2 - The method. VAEs, CVAEs, and what a shared latent space actually buys you.

Act 3 - Does it work? (replication). Can 2 latent dimensions beat 3 NS factors on our data?

Act 4 - So what? (extensions). Two trading strategies, a carry sanity check, and a cross-currency signal that *only* the shared latent space can produce.

One-sentence summary
A 2D neural compression does beat the classical 3D basis - and the residuals turn out to be tradable.

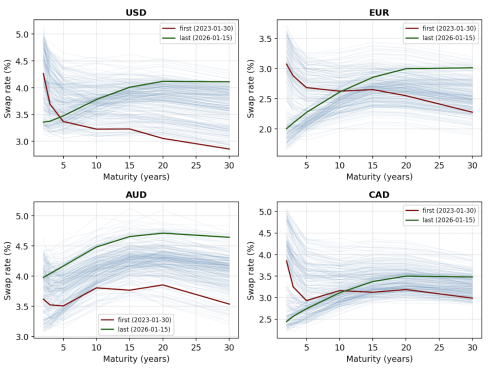
The yield curve, in one slide

What it is. For each currency, on each day, swap rates quoted at maturities 2Y, 3Y, 5Y, 10Y, 15Y, 20Y, 30Y. Rate vs maturity ⇒ a **curve**. Summarises expectations about rates, inflation, policy.

Why we care.

- Every fixed-income derivative is priced off it
- Risk management needs to simulate it
- Central banks watch it for signals

The technical problem. The curve is 7 numbers. We want to work with **fewer** (2 or 3) that still capture its shape. This is *compression*.



Each faint line = one day's curve. Regimes visible: inversions (2023), normalisation (2024–25), JPY flat & low.

Why compression matters - The trade-off

Too many state variables

- Slow Monte Carlo
- Overfit noise
- Hard to calibrate
- Unstable in stress

Too few state variables

- Miss genuine curve shapes
- Can't price butterflies
- Can't simulate inversions
- Can't generalise across regimes

The question: what is the **minimum** *number of numbers that still describes a real curve?*

- Classical answer (since 1987): **three**, via Nelson-Siegel
- Sokol's (2022) answer: **two**, via a VAE

The classical answer: Nelson-Siegel (1987)

Three factors with transparent economic meaning:

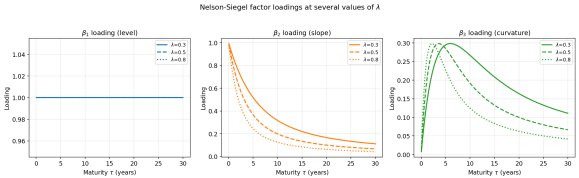
$$R(\tau) = \underbrace{\beta_1}_{\text{level}} + \underbrace{\beta_2 \frac{1-e^{-\lambda\tau}}{\lambda\tau}}_{\text{slope}} + \underbrace{\beta_3 \left(\frac{1-e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right)}_{\text{curvature hump}}$$

Why it dominates.

- Closed-form OLS once λ is fixed
- Interpretable: level / slope / curvature
- Central-bank standard for 35+ years

Why it's not optimal.

- Born from Taylor expansion of a 2-exponential model, *not* from minimising reconstruction error
- Large parts of $(\beta_1, \beta_2, \beta_3)$ space produce curves *never* observed
- One fit per currency, no cross-market sharing



Level (flat), slope (decaying), curvature (hump) at $\lambda^\star = 0.500$.

Sokol's bet - and our research questions

Sokol's geometric insight. Real curves live on a low-dimensional *manifold* inside $\mathbb{R}^7$. NS imposes three exponential axes onto that manifold. A neural encoder can learn the manifold *directly from data*.

The payoff, if it works.

1. Fewer dimensions for the same accuracy (2 beats 3)
2. Every decoded point is a *realistic* curve by construction
3. Cross-currency information sharing $\Rightarrow$ data-scarce markets benefit from data-rich ones

What we set out to answer

Q1. Can a 2D CVAE actually beat 3D Nelson-Siegel on our data?

Q2. Does the learned latent space have economic meaning?

Q3. If the reconstruction is better, are the *residuals* tradable?

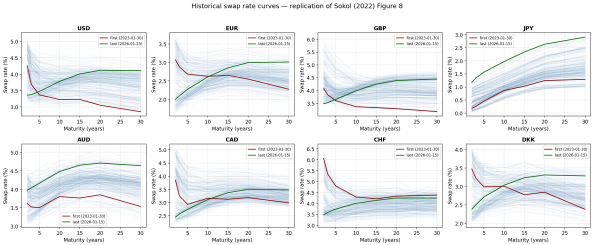
The data

Bloomberg OIS swap rates

- **7 currencies**: AUD, CAD, DKK, EUR, GBP, JPY, USD
- **7 tenors**: 2, 3, 5, 10, 15, 20, 30 Y
- **Daily**, 30 Jan 2023 → 15 Jan 2026
- **774 business days** per currency
- Train / OOS split at **31 Mar 2025** (≈ 70/30)
- **208 OOS business days** for trading tests

CHF dropped: Bloomberg series we had predates SARON (quoted 2Y at 3–6% vs true 0.5–1.8%). Including it would poison the shared latent space.

Caveat. Sokol: 13 ccy, 20+ yr. Us: 7 ccy, 3 yr. Smaller, but rich in regimes.



Per-currency historical curves. First/last day highlighted. JPY lives in its own world.

VAE in one slide

Encoder. $\mathbf{S} \in \mathbb{R}^7 \;\mapsto\; (\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2)$

Map a curve to a *distribution* over a K-dim latent space.

Sample (reparameterisation trick):

$$z_k = \mu_k + \sigma_k \, \omega_k, \quad \omega_k \sim \mathcal{N}(0,1)$$

Decoder. $\mathbf{z} \in \mathbb{R}^K \;\mapsto\; \mathbf{S}' \in \mathbb{R}^7$

Loss (Sokol, Eq. 5):

$$\mathcal{L} = \frac{1}{|B|} \sum_{i \in B} \sum_{n=1}^{7} (S'_{i,n} - S_{i,n})^2 + \beta \cdot D_{\mathrm{KL}}$$

with $\beta = 10^{-7}$ reconstruction dominates, KL only prevents gaps.

Why distributions, not points?

A plain autoencoder leaves **holes** in latent space. Forcing the encoder to output a Gaussian ball around each curve, and training the decoder to handle *any* sample from that ball, fills the gaps. $\Rightarrow$ **dense, gap-free manifold**, every point decodes to a plausible curve.

Why force $K = 2$?

If $K \geq 3$, any win would be confounded with having *more* parameters than NS. We force the VAE to beat NS with *fewer* dimensions or lose honestly.

Three architectures, same latent dimension $K = 2$

Single-currency VAE

99 params, one model per ccy

Layer	In	Out
Enc	7	4
Enc	4	4
Dec	2	4
Dec	4	7

Baseline.

7 separate models, no sharing. Our stress test.

Multi-currency VAE

191 params, one model for all

Layer	In	Out
Enc	7	7
Enc	7	4
Dec	2	4
Dec	4	7
Dec	7	7

Shared map.

One latent space, no currency label. Crucial for Ext 2.

CVAE

433 params, conditioned

Layer	In	Out
Enc	7+7	13
Enc	13	4
Dec	2+7	7
Dec	7	7
Dec	7	7

Best fit.

One-hot currency concatenated to encoder & decoder.

All three: $K = 2$, Tanh hidden, Sigmoid output, Adam, 5 000 epochs, $\beta = 10^{-7}$. Deliberately **tiny** the point is compression, not capacity.

Nelson-Siegel benchmark: cross-validating λ

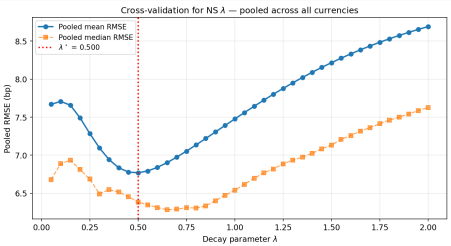
We refuse to hard-code Diebold-Li's $\lambda \approx 0.73$.
We pick $\lambda^\star$ by **pooled cross-validation**:

$$\lambda^\star = \arg\min_{\lambda} \ \overline{\mathrm{RMSE}}_{\mathrm{pooled}}(\lambda)$$

Result. $\lambda^\star = 0.500 \Rightarrow$ curvature hump at $\tau^\star = 2\,\mathrm{yr}$.

Smaller than the literature value. Why? Our sample is dominated by **short-end features**: aggressive 2023 inversions at 2Y–5Y.

Fair comparison. Multi-VAE shares parameters across currencies; NS should too. Hence one *global* $\lambda^\star$.



Pooled RMSE vs λ. Unimodal. Minimum at $\lambda^\star = 0.500$.

NS benchmark - per-currency RMSE (bp)

Model	AUD	CAD	DKK	EUR	GBP	JPY	USD	**Pooled**
NS (3D, $\lambda^\star$)	10.00	7.43	8.31	6.35	2.53	6.49	6.28	**6.77**
eNS (3D, λ per date)	8.90	6.77	7.73	5.82	2.36	2.59	5.21	**5.62**
NSS (4D, ceiling)	2.03	2.25	2.04	1.02	0.45	1.53	1.32	**1.52**

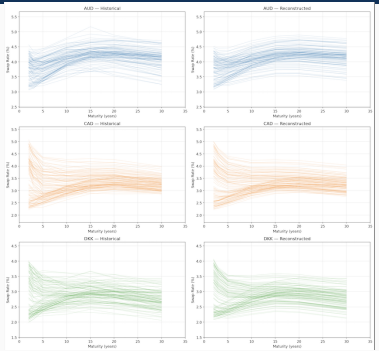
- NS target to beat: 6.77 bp pooled. This is what 2D VAE must clear.
- AUD is the hardest (10.00 bp) — its 30Y point refuses the NS shape.
- Biggest NS $\rightarrow$ eNS gain: **JPY** (6.49 $\rightarrow$ 2.59). Global $\lambda^\star$ is badly adapted to JPY's flat, near-zero curve.
- NSS (4D) is a ceiling, not a competitor. 4D vs our 2D is unfair, we use it as sanity check only.

THE main result: 2D CVAE beats 3D Nelson-Siegel

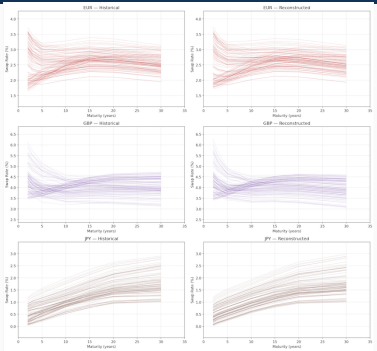
Model	AUD	CAD	DKK	EUR	GBP	JPY	USD	**Pooled**
NS (3D)	10.00	7.43	8.31	6.35	2.53	6.49	6.28	**6.77**
NSS (4D) ceiling	2.03	2.25	2.04	1.02	0.45	1.53	1.32	**1.52**
Single-VAE (2D)	12.97	11.85	15.75	13.20	2.48	4.99	8.11	9.90
Multi-VAE (2D)	5.77	6.10	6.28	3.38	6.78	4.63	5.52	5.49
CVAE (2D)	**4.00**	**4.53**	**4.88**	**2.75**	**2.49**	**3.00**	**4.64**	**3.80**

- **CVAE: 3.80 bp pooled at $K = 2$, vs NS 6.77 bp at $K = 3$.**
- **CVAE beats NS on all 7 currencies.** No exception.
- Multi-VAE (no conditioning) beats NS on 6/7 - GBP is the one failure: the shared space sacrifices GBP's easy shape to accommodate JPY.
- Single-VAE *loses* to NS on 5/7 - posterior collapse on small data. **Joint multi-currency training is mandatory at this compression.**

Do the reconstructions look realistic?



AUD, CAD, DKK: inversions and steepening well captured.



EUR, GBP, JPY: different regimes, including JPY low-rate structure.

Key takeaway. The CVAE does not just reduce RMSE: it reproduces realistic yield-curve shapes across currencies and regimes.

Only visible bias: slight smoothing of extreme curves toward the mean, consistent with VAE regularisation.

Statistical significance

Wilcoxon signed-rank on paired daily RMSE differences (CVAE vs NS). H_0: median difference $= 0$.

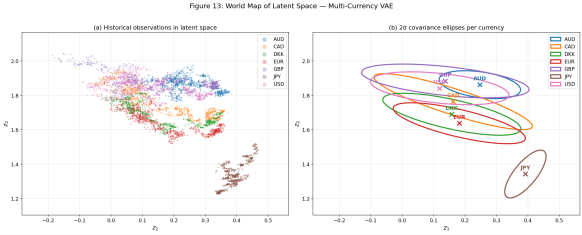
Currency	$\overline{\Delta}$	p-value	Winner
AUD	$+6.00$	$< 10^{-16}$	CVAE
CAD	$+2.89$	$< 10^{-16}$	CVAE
DKK	$+3.43$	$< 10^{-16}$	CVAE
EUR	$+3.59$	$< 10^{-16}$	CVAE
GBP	$+0.04$	10^{-5}	CVAE
JPY	$+3.49$	$< 10^{-16}$	CVAE
USD	$+1.63$	$< 10^{-16}$	CVAE
Pooled	$+3.01$	$\ll 10^{-16}$	**CVAE**

- CVAE beats NS **systematically** across all currencies
- Economic significance varies: **large gains** (AUD, EUR, JPY), **negligible for GBP**

Very small p-values reflect large sample size and persistent differences; they should be interpreted with caution due to time dependence.

Latent world map - economic meaning emerges

The model is never told the currency — yet it discovers economic structure.

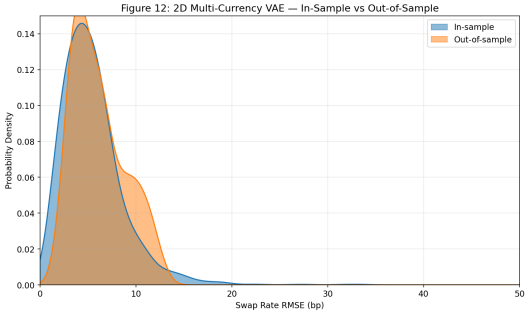


- **JPY island** at $z_2 \approx 1.34$ → BoJ low-rate regime
- **EUR ≈ DKK** → currency peg recovered from data
- **Western cluster** → synchronised monetary cycle
- z_2 = **level**, z_1 = **shape**

This is the key qualitative result.

The latent space has real economic meaning and enables cross-currency signals.

Out-of-sample: does it overfit?



Retrain Multi-VAE on 70% split. Test on 30% OOS.

- In-sample RMSE: 5.45 bp
- Out-of-sample: 6.12 bp
- Ratio: $1.12\times \ll 1.5$

Why no catastrophic overfit?

- Tiny param count (191 vs 3,962 obs, 21:1)
- Sigmoid-capped output bounds predictions
- Joint training across 7 ccy is itself a regulariser

From compression to trading — why bother?

So far: the CVAE fits the curve better than NS. Does that translate into **tradable signals**?

The trading idea. The CVAE reconstructs a "fair-value" curve. When the *actual* curve diverges from this reconstruction, two things can happen:

1. A genuine new feature the model hasn't captured $\rightarrow$ *hold off*
2. Transient noise on top of stable dynamics $\rightarrow$ **mean reversion trade**

If the model is good, the residual is mostly case (2). That's the bet.

What we will test
Strategy A. 2s10s slope residual
Strategy B. 2s5s10s butterfly residual (curvature)
Ext 1. Carry decomposition, is the alpha real?
Ext 2. Cross-currency, only possible with the shared latent space

Strategy A: 2s10s slope residual

Signal.

$$\Delta_t^{\text{slope}} = \underbrace{(S_{10Y} - S_{2Y})(t)}_{\text{market}} - \underbrace{(S'_{10Y} - S'_{2Y})(t)}_{\text{CVAE fair-value}}$$

60-day rolling z-score $\Rightarrow$ enter $|z| > 1$, exit $|z| < 0.3$, 0.5 bp t-cost/flip.

Signal	**Sharpe**	Total (bp)	Ann. (bp)	MaxDD (bp)	HitRate
CVAE_resid	+3.99	+79.4	+96.2	−5.9	0.56
Naive (raw slope z)	+2.18	+41.3	+50.1	−13.3	0.54
NS_resid	−2.35	−32.8	−24.8	−31.1	0.46

The headline lesson of the project

NS fits the curve *well* (6.77 bp RMSE) but its residuals give Sharpe −2.35. Same logic on CVAE residuals: +3.99. **A good fitting model is not automatically a good signal generator.**

Strategy B: 2s5s10s butterfly (curvature)

Butterfly. $b_t = 2 \cdot S_{5Y,t} - S_{2Y,t} - S_{10Y,t} \rightarrow$ DV01-neutral to parallel shifts and slope. Pure curvature. The 5Y belly is where the NS exponential struggles most.

Strategy	**Sharpe**	MaxDD (bp)	HitRate
CVAE butterfly	+5.12	−7.5	0.61
NS butterfly	+3.36	−9.7	0.54
Naive butterfly (z of raw)	+5.39	−3.3	0.50

Honest result.

- vs NS: CVAE wins clearly (+5.12 vs +3.36) $\Rightarrow$ data-driven basis helps
- vs Naive: **we lose by 0.27 Sharpe.** On this 9-month window, post-COVID normalisation made raw butterflies trivially mean-reverting; no model was needed.
- *On a longer / noisier window we'd expect the Naive signal to break on regime shifts; we cannot claim it from 208 days.*

Extension 1: is the alpha real, or just carry?

The worry. A 2s10s trade earns daily *carry* regardless of rate moves. On an inverted curve (2023–24), steepener carry is positive. Is Strategy A just a disguised carry trade?

Decompose. $\mathrm{pnl}_t = \underbrace{(\mathrm{pnl}_t - \mathrm{carry}_t)}_{\text{pure alpha}} + \underbrace{\mathrm{carry}_t}_{\text{carry}}$

Answer

Portfolio: pure-α Sharpe 1.81 $>$ total Sharpe 1.69. The alpha *survives* a carry headwind of -74% on average. **Alpha is genuine, not a carry trade.**

JPY caveat: its contribution is almost *entirely* carry.

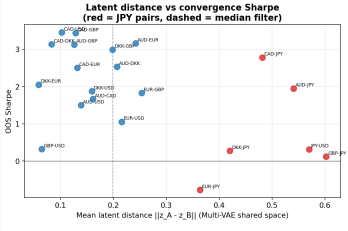
	AUD	CAD	DKK	EUR	GBP	JPY	USD	**Portfolio**
Sh. total	+3.77	+2.79	+2.72	+1.69	+0.10	+0.58	+0.18	+1.69
Sh. alpha	+3.69	+2.75	+3.24	+2.57	+0.17	+0.05	+0.18	+1.81
Sh. carry	+1.85	+0.56	−7.43	−7.38	−1.15	+4.81	−0.04	−1.25

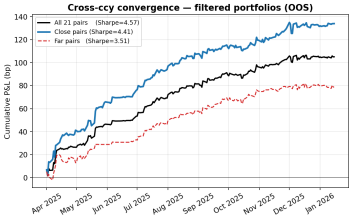
Extension 2: cross-currency latent convergence

Only possible with the shared latent space. NS factors and CVAE latents are per-currency. Only Multi-VAE gives *one* map for all 7.

Trade. For each pair (A, B): z-score slope spread $s_t^{AB} = \text{slope}_t^A - \text{slope}_t^B$. Mean-revert. Use latent distance $\|z_A - z_B\|$ as a **pair-selection filter**.

Portfolio	Sharpe
All 21 pairs	+4.57
Close pairs (bot. 50%)	+4.41
Far pairs (top 50%)	+3.51
JPY pairs (mean)	+0.78
non-JPY pairs (mean)	+2.31





Non-JPY Sharpe is 3× JPY pairs. JPY vs Western is a *structural* divergence (BoJ policy), not transient — and the latent distance identifies it. **The latent geometry has real economic content.**

Honest caveats

- **Short OOS window.** 208 days $\approx$ 0.83 yr. Sharpe SE $\approx$ 1.1. Bonferroni-corrected significance threshold $\mathrm{Sharpe} \gtrsim 2.9$. Portfolio Sharpes 3.99 and 5.12 clear it; per-currency numbers below $\approx$ 2.1 are rankings, not findings.
- **Butterfly honesty.** Strategy B lost to the Naive butterfly by 0.27 Sharpe. Post-COVID normalisation made butterflies trivially mean-reverting $\rightarrow$ longer sample needed to tell if CVAE adds value.
- **JPY in Strategy A is mostly carry.** $\mathrm{Sharpe}_{\alpha} = 0.05$ vs $\mathrm{Sharpe}_{\mathrm{carry}} = 4.81$. The JPY contribution is not model-driven.
- **Only 7 currencies.** Sokol had 13. Conclusions on posterior collapse and on the Multi-VAE's GBP failure depend on that.

Take-aways

Replication

- **2D CVAE $\geq$ 3D NS on every one of 7 currencies** ($p < 10^{-16}$ pooled, $\overline{\Delta} = +3.01$ bp)
- Multi-VAE organises the latent space economically *without supervision* - JPY isolation, EUR–DKK overlap, Western cluster all emerge from data
- Single-currency VAE collapses at $K = 2$ on short data $\Rightarrow$ joint multi-currency training is mandatory at aggressive compression

Trading extensions

- Strategy A: CVAE Sharpe $+3.99$ vs NS -2.35 - fitting $\neq$ signal quality
- Strategy A alpha survives carry: pure-α Sharpe 1.81 $>$ total 1.69
- Strategy B: wins vs NS (5.12 vs 3.36), loses vs Naive (5.12 vs 5.39) on this window
- Extension 2: non-JPY pairs Sharpe 2.31 vs JPY 0.78 $\Rightarrow$ latent geometry has economic meaning

One-line conclusion

A 2D neural compression beats the 3D classical basis and the residuals are tradable.

Thank you

Questions?

Code and data available on GitHub:

https://github.com/hamzaelarji/topics-in-derivative-pricing-Coursework